Chiharu Mamiya

Stockholm, Sweden | linkedin.com/in/chiharumamiya | +46 727653690 | chiharu.mamiya38@gmail.com

Professional Summary

Power systems engineer with experience in FACTS support at GE Vernova and PSS/E automation at Siemens. Specializing in grid simulation, transmission-level power system analysis, and Python-driven workflow acceleration. Currently pursuing the EIT InnoEnergy SENSE double-degree MSc at KTH and KU Leuven, with focus on HVDC transmission, multi-terminal grid interoperability, and renewable integration.

Education

Master of Science **KTH Royal Institute of Technology** *Stockholm, Sweden*

- Major in Smart Electrical Networks and Systems (EIT InnoEnergy SENSE); Year 2 at KU Leuven
- Coursework: Electricity Pricing and Emissions, Power System Analysis, Power Electronics, Power Grid Technology and Substation Design
- Honors: InnoEnergy Diversity, Equity,and Inclusion Scholar

Bachelor of Science **Union College** *Schenectady, NY*

- Major in Electrical Engineering, Minor in Biomedical Engineering, GPA: 3.8
- Honors: Tau Beta Pi Engineering Honor Society, JASSO Scholar, Grew Bancroft Scholar, UAS7 Scholar, NYSTAR Scholar

Publications

Mamiya, C. A., Marletta, L. M., Traver, C., Cotter, S., Okwori, M. (2025). *Predictive Analysis of Basketball Shots from Inertial Data Using Machine Learning.* IEEE AIIoT 2025. ieeexplore.ieee.org/document/11105371

Engineering Experience

Grid Solutions Graduate Engineer **GE Vernova** *Saratoga Springs, NY* **10/2025 – 05/2026**

- Performed short-circuit, load flow, and steady-state studies for HVDC-interconnected and FACTS-integrated transmission networks in PowerFactory, supporting industrial and utility grid modernization studies.
- Developed Python automation for PowerFactory workflows, accelerating short-circuit studies and data analysis.
- Aided in developing FACTSFLEX product interface (HTML, CSS, Django) used by the presales team in customer-facing presentations.

Technical Consultant Intern **Siemens** *Fremont, CA* **06/2025 – 08/2025**

- Enhanced ACCC Python tools to streamline contingency and power flow analysis in PSS/E, increasing efficiency by up to 70%.
- Optimized TSAT plotting tool runtime by 94% through code refactoring and parallel processing using optimized libraries, enhancing simulation performance.
- Developed test suites and process flowcharts in Microsoft Visio to support debugging, regression testing, and onboarding, improving overall code reliability.

Software Developer Intern **Siemens** *Schenectady, NY* **06/2024 – 08/2024**

- Automated XML data extraction via Python scripts, reducing manual workload by 25% and pushing updates to GitHub.
- Migrated legacy C++ modules to MFC, enhancing cross-platform integration across x86-based infrastructure.
- Communicated progress in daily stand-ups, contributing to 2-week early project delivery.

Projects

Nordic BESS Revenue-Stacking Model, *ENTSO-E & Svenska kraftnät data* **09/2026 – Present**

- Modeling a 10 MW battery in SE3 across FCR-D, FCR-N, and day-ahead arbitrage with a project-finance layer (debt sizing, DSCR, equity IRR). Analyzes FCR-D price compression 2023–2025 and its implications for BESS bankability.

N-1 Contingency Screening, *IEEE 39-Bus System data* **06/2026 – 07/2026**

- Automated 35-contingency N-1 screening per NERC TPL-001 principles. Identified critical line overloads (max 157.5% loading) and quantified how distributed wind integration reduced violations by up to 77% across five penetration scenarios.

Skills

- **Power systems:** PSS/E, PowerFactory, TSAT, pandapower, contingency & power-flow analysis, short-circuit studies
- **Programming & data:** Python (pandas, NumPy, scikit-learn), Git/GitHub, MATLAB/Simulink, SQL basics, C/C++
- **Markets & finance (developing):** Nord Pool / ENTSO-E data analysis, LCOE & project cash-flow modeling, DSCR/IRR analysis
- **Languages:** English (native) · Japanese (native) · German (B1 in progress) · Swedish (beginner) · Mandarin (elementary)